

Manual

Archiving Shop Floor / Order Data BDE-ARC 8.2

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1 Archiving of Shop Floor / Order Data

Purpose

The function package "archiving shop floor /order data" makes it possible to access archived data and, moreover, to export recorded data and to load these data back into the system as needed.

Implementation notes

You use the function package "archiving of shop floor / order data", if:

- You would like to have access to data already moved to archive tables in the function package "shop floor data collection".
- Because of legal requirements or customer demands, you need to ensure that the data entered is stored long-term.
- Because of legal requirements or customer demands, you need to import data that has already been exported back into the system to be evaluated / analyzed again.

Integration

Archiving of shop floor / order data refers to inventory data and recorded data of archive tables pertaining to the function package Shop Floor Data Collection.

Features

- Direct access to archive tables from the application
 - Direct access to archived shop floor / order data from the applications
 - Order overview
 - Order information
 - Order shift log
 - Personnel shift log
 - Personnel report
 - Schedule controlling
 - Overhead cost controlling
 - Maintenance controlling
 - Production controlling
 - Scrap statistics
 - Scrap profile
- Transport function
 - Functions transferring data from online tables into archive tables.
- Export function

- Functions used to transfer (export) data from archive tables to external file systems for the purpose of storing the recorded shop floor / order data long-term.
- Import function
 - Functions importing exported data into archive tables in order to evaluate this data using the applications listed above.

2 BDE-Specific Configurations

Overview

In the BDE module, by default data are held in cache for 35 days before they are moved into long-term storage.

In a variety of BDE reports, there is the option to pull up data that are 35 days old or older. To do this, the BDE postings are set in a special medium-term or archive area. You automatically have access to such data for the most part if the selection period exceeds the short-time data area. In some applications, there is the option to "Consider long-term data" in the selection area, which can be accessed from here.

The data considered during BDE archiving include:

- Order backlog
- Order status
- Order actions
- Order sequences
- Order networks
- Lists of material components and production resources and tools
- Order serial numbers
- Additional information (long texts)
- Order quantities
- Additional order information
- Order posting records
- Order backlog (PPS)
- Additional order information (PPS)
- Order actions (PPS)
- Master detail user field (specific table)
- Order specific events
- Personal events
- Additional information for events

- Order logging information
- Additional order logging information
- BDE configuration logging information
- Additional BDE configuration logging information

Configuration

You can use HYDRA data management to configure the retention period for the data in each of the data areas.

When transferring data into the archive tables, the data for which the "retention period" (in number of days/ months/ years; see the values in parentheses) has been exceeded is transferred. If the archiving license relevant to the BDE is not available, the data will be deleted after the set retention period.

Product	Object	Object designation	Transfer	Factory default interval
BDE	ANR	Order backlog -Status, - Actions, - Sequences, - Networks, - Lists of material, - Serial numbers, - Additional information, - Quantities, - PPS data, - Master detail user fields	Online data → medium-term archive	35 days
BDE	A_ANR	Long-term archiving: Order backlog -Status, - Actions, - Sequences, - Networks, - Lists of material, - Serial numbers, - Additional information, - Quantities, - PPS data, - Master detail user fields	Medium-term archive: → long-term archive	2 years
BDE	ADEPRO	Posting records of orders	Online data → medium-term archive	35 days 1)
BDE	ADEPRO_ADD	Posting record of orders – additional information	Online data → data is deleted (it is not transferred to the medium-	Delete if the relevant posting record of the

Product	Object	Object designation	Transfer	Factory default interval
			term data)	order is no longer available
BDE	A_ADEPRO	Long-term archiving: Posting records of orders	Medium-term archive → long-term archive	2 years 1)
BDE	EREIGADEA	Order-related events incl. additional information	Online data → medium-term archive	35 days 2)
BDE	A_EREIGADEA	Long-term archiving: Order-related events incl. additional information	Medium-term archive: → long-term archive	2 years 2)
BDE	EREIGADEP	Personal events incl. additional information	Online data → medium-term archive	35 days 2)
BDE	A_EREIGADEP	Long-term archiving: Personal events incl. additional information	Medium-term archive: → long-term archive	2 years 2)
BDE	ANRLOG	HYDRA logging data	Online data → medium-term archive	35 days
BDE	A_ANRLOG	Long-term archiving HYDRA logging data	Medium-term archive: → long-term archive	3 years
BDE	CFGLOG	HYDRA logging configuration	Online data → medium-term archive	35 days
BDE	A_CFGLOG	Long-term archiving HYDRA logging configuration	Medium-term archive: → long-term archive	3 years

Please note:

- 1) If the values entered for ADEPRO or A_ADEPRO are changed (increased), the entries for ANR or A_ANR will also have to be changed (increased) accordingly.

Provided that the BDE log records are to be archived at the earliest after the OP has been archived, this can be achieved by defining the following condition for the object ADEPRO in the field of the same name within the "data management" configuration:

ade_protokoll.auftrag_nr in (select auftrag_nr from a_auftrag_status)

Please note that the [order-related postings](#) only allow for data of the online data area to be selected and edited.

- 2) Please note that the [event maintenance](#) only allows for data of the online data area to be selected and edited.

3 Data Management

Overview

Menu	System administration → Archiving → Data management
Transaction code	arccfg
Function authorization	arccfg.*

Purpose

Use this application to view or change the centrally managed [Archiving](#) settings for recorded data.

Integration

Use this application to configure the settings for archiving / data management for all components/functions.

Field descriptions

Product

Enter the HYDRA product group for which you want to define a rule in the *Product* field.

Object

Use the *Object* to define the data you want to retain.

Retention period

Use the fields for the *Retention period* to define how long data should be available until data is archived.

Unit

Specify the retention period in days, months or years.

Last retention date

This field is computed and specifies the last day when data is still available in this object.

Action

You can choose from three options to specify the processing for an object:

D = delete	Object is deleted
M = move (archive)	Object is transferred to the next area
X = export	Object is unloaded (XUNLOAD format) and then deleted

Target object

Currently, the target object automatically results from the detail configuration.

Condition

The characters in this field are added as an additional condition to the database command that controls the action. This condition is linked to the other conditions of the original command using AND and is set in parentheses.

Last run (date/time)

Indicates the point in time when this rule was applied last.

License

Indicates which license is required for archiving. You can enter several licenses (separated by a space). If none of these licenses is licensed in the system, the data is either deleted (action D) or unloaded into files (action M or X), depending on the relevant action.

Path

Optional path to generate the file export (unload). The archiver currently only supports local HYDRA server drives. If you do not enter a path, file exports are filed as follows:

<HYDRADIR>/<SYSTEM>/custom/archive/<YYYY-MM-DD>/<PRODUCT>/

HYDRADIR ... HYDRA directory

SYSTEM ... System number

YYYY-MM-DD ... Archiving date (YYYY ... Year, MM ... Month, DD ... Day)

PRODUCT ... Product group from archiving configuration

Administration table

Name of the administration table where archiving logs are stored.

Administration duration (retention period of administration table)

Use the fields for the *retention period* to specify how long the logs should be available in the administration table before being deleted.

Unit

Specify the retention period in days, months or years.

Archiving step

Specifies if this archiving process uses archiving function I or II. Archiving function I: The function moves data from the online data set to archive tables or deletes the data: setting M (medium-term archive). Archiving function II: The function moves data from the archive table to the file export or deletes the data: setting L (long-term archive).

Configuration

Indicates whether or not the configuration is active. Possible values: Y/N

Archiving type

Identifier for time or object-related archiving. Supported modes: O = Object-related archiving (i.e. data is archived for each object individually). Z = Time-related archiving (i.e. data is archived without any object reference).

Note: Time and object-related archiving differ from each other significantly in the archiving performance (runtime). Object-related archiving of mass data is not recommended.

Priority

Integral value greater than 0. Indicates the processing sequence if several objects are defined for a product group. Processing starts with the lowest value.

Master table

Table including the data to be archived. The extensions entered in the *Condition* field refer to this table.

Date column

Date column in the master table. The system uses this date column to evaluate the retention period for the data to be archived.

Key 1

Unique key column in the master table; this key identifies the data to be archived. You can define up to 5 key columns for object-related archiving. Time-related archiving only supports one primary key in *Key 1*.

Keys 2 – 5

Additional optional key columns for object-related archiving.

Comment

Use the comment line to describe the archiving configuration.



If you copy the archiving configuration, the system currently only copies the data management configuration, but no defined data records from the object details. Currently, you must use the respective dialog to copy the data records of the object details to the new configuration.

Module	Version(s)	Description of archiving
BDE	8.1	See here
CAQ	8.1	See here
LLE	8.1	See here
MDE	8.1	See here
MPL	8.1	See here

PEP	8.1/8.2	See here
PZE/PZW	8.1/8.2	See here
ZKS	8.1/8.2	See here

4 Reload Manager

Summary

HYDRA menu	System administration → Archiving → Reload manager
FEDRA menu	System administration → Archiving → Reload manager
Transaction code	arcrlid
Function authorization	arcrlid arcrlid.export arcrlid.import

Purpose

The customer is responsible for backing up data in a data archive and for restoring data. When data is backed up it is transferred from the long-term data area to the archive data area and, as a result, it is stored in a separate file system. The customer can start the backup process using the Reload Manager.

When data, which is filed in the archive data area, is restored, it is transferred back (copied) to the long-term data area. The customer has to do this and bears the responsibility.

To be able to evaluate restored data using standard evaluations/reports, the data has to be transferred back (copied) to the reload data area.

For this reason, the Reload Manager enables the following functions:

1. Moving of exported data into the customer archive (the customer archive path has to be defined within the HYDRA path settings).
2. Loading of exported data into the reload area for evaluations

Integration

The Reload Manager is a central function that is used by many components or functions.

Selection Criteria

The application provides the following selection criteria:

Module

Reference to the product group which archived data belongs to.

Object type

Object type of the archived data.



The selection options “module” and “object type” are mandatory fields.

Toolbar

Export

The “EXPORT” button allows for the entries selected in the Reload Manager to be exported to the customer directory (all data records can be selected using the context menu of the right mouse button). Once the function has been started, the input dialog that opens requires the customer archive path to be entered.

Import

The “IMPORT” button allows for the selected entries to be loaded into the reload area, in order for the data to be again available for HYDRA evaluations/reports. Once the function has been started, the data loading mode and an optional path are requested. By indicating the path, it is possible to specify another storage location for the files, provided that the archived files have not been moved to the customer-specific archive using the Reload Manager.

There are three different ways to deal with reload data after they have been reloaded to prevent the reload data set from increasing excessively (→ no slow evaluations/reports):

The following modes are distinguished:

- **Cyclic**
Data is loaded to the reload area in the “cyclic” mode. Loaded data is automatically removed from the corresponding reload tables, once the retention period specified within archive settings has expired.
Please compare the configuration entry HYD / RELOADMANAGER of the data management function. When demo settings are used, data is removed from the reload area after 14 days.

When data is imported, the `HYD_REL_MANAGEMENT.DELETE_DATE` field is calculated subject to the configuration and entered in the reload management table.

- **User-specific**
Data is loaded to the reload area in the “user-specific” mode. The loaded data is automatically removed from the corresponding reload area, once the time specified for the user within the user-specific settings has expired. Exception: In case several users loaded identical data into the reload area, the corresponding retention period is automatically set to the maximum date.
When data is imported, the `HYD_REL_MANAGEMENT.DELETE_DATE` field is computed subject to the configuration and entered in the reload management table.

- Manual

Data is loaded in the “manual” mode to the reload area. The customer is responsible for deleting data from corresponding reload tables. However, identical data cannot be loaded several times in the “manual” mode (data can be loaded in the “user-specific” mode though).

In order for data to be transferred to the reload data area, the HYDRA server must be able to access these files. Otherwise, loading is cancelled with an error message.